CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Plan a small casting without pouring

Goal: I can draw a useful small object, estimate its size, and compare possible materials.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 9, on-track choice

Ask | Plan | Build | Test | Improve

You are done when

Show a labeled design and explain why your chosen material fits its use.

You will use

- Paper
- Pencils
- Rulers
- Calculators if available
- Empty dry mold or dimensioned drawing
- Teacher-cleared cured example or picture

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Look at a dry mold or its measured drawing.
- Identify the inside space that shapes the object.
- Compare a cardboard model with a teacher-cleared cured example or picture.

4 Step 4: Make it work — 17 min

- Sketch a small token, tag, or game piece.
- Measure a rectangular practice cavity or its drawing.
- Use length, width, and depth to estimate its volume.

5 Step 5: Check one thing — 5 min

- Compare making your design from cardboard and casting material.
- Name one reason for each choice.
- Improve the design using the teacher's size limit.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show a labeled design and explain why your chosen material fits its use.

What to turn in

Your labeled drawing and size estimate.

No computer? Another way to do it

The whole lesson is paper based.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

This lesson stays dry. Students do not mix, pour, heat, demold, or handle uncured resin.

Grade targets

Grade 5: Teacher-supported measuring and a labeled drawing.
Grade 6: Calculate the volume and explain an irregular mold.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.